

Nagarjun Mallesh

nagarjunmallesh@gmail.com | 857-799-0214| Boston, MA | [linkedin.com/in/nagarjun-mallesh](https://www.linkedin.com/in/nagarjun-mallesh) | github.com/NagarjunMa

SUMMARY

FullStack Software Engineer with 3+ years building **scalable applications**, backed by a Master's in Information Systems and a Bachelor's in Information Science & Engineering. Passionate about **AI**, system optimization, and **solving complex problems** with **future-scalable architectures** following **AWS Well-Architected Framework** principles.

EXPERIENCE

Jul 2025 - Present **Systems Analyst** **ML Technologies LLC, US**

- Created **vehicle recovery system** powered by **AWS Bedrock Knowledge Base** with **S3** and **OpenSearch**, improving emergency response coordination by 70%.
- Designed an emergency services platform using **API Gateway**, **Lambda**, and **Step Functions** for automated categorization, streamlining response workflows by 60%.

Jul 2024 - Jul 2025 **Backend Software Engineer** **Keelworks Foundation, US**

- Designed serverless-first platform architecture** adhering to AWS Well-Architected Framework pillars (operational excellence, security, reliability, performance efficiency, cost optimization), implementing EventBridge-Lambda workflows that **reduced infrastructure maintenance overhead by 60%**.
- Implemented RAGAS framework solutions** to reduce hallucinations and mitigate semantic entropy in generative AI models using advanced evaluation metrics and guardrails, **improving AI response accuracy by 45% and reducing false favorable rates**.
- Transformed deployment architecture** through Docker containerization and GitHub Actions automation with AWS ECR, implementing blue-green deployment strategies that **reduced deployment cycles from hours to minutes, with 95% error reduction**.
- Architected an intelligent onboarding assistant** using AWS Bedrock RAG pipeline with Knowledge Base and Amazon Nova Pro LLM, implementing document parsing and semantic search capabilities that **reduced volunteer onboarding time by 70%**.

Oct 2019 - Jul 2022 **Software Engineer** **Hewlett-Packard, IN**

- Built a responsive management platform** using React.js with Redux state management, implementing custom hooks and context API for efficient state handling across 50+ components, **improving application load times by 35%**.
- Architected robust backend services** for management platform using NodeJS and MongoDB, designing custom workload algorithms and efficient data processing pipelines to handle 10,000+ concurrent operations, **ensuring 99.9% uptime and seamless user experience**.
- Developed BIOS configuration system** using PowerShell and VBS scripting with automated validation workflows, **transforming week-long manual processes into 15-minute computerized operations**.
- Architected cloud-based OS deployment pipeline** using AWS services and C# automation scripts, implementing parallel processing and error recovery mechanisms, **reducing provisioning time from days to hours, and achieving a 98% success rate**.

Jul 2019 - Oct 2019 **Associate Software Engineer,** **Digital API Craft Pvt Ltd, IN**

- Spearheaded migration from monolithic to microservices architecture** for legacy banking system using Java Spring Boot, decomposing application into independent microservices with Kafka messaging integration, **reducing system complexity and improving deployment frequency by 300%**.
- Engineered user services and transaction management microservices** using Spring Boot REST APIs with distributed transaction patterns, implementing event-driven architecture with Redis caching, **handling 50,000+ daily transactions with 99.8% reliability**.
- Conceptualized user profile management system** with domain generation capabilities, utilizing Golang APIs for DNS pool integration and domain validation based on networking fundamentals, **automating domain provisioning, and reducing manual errors by 90%**.

SKILLS

- Python	- SpringBoot	- JavaScript / TypeScript
- Java	- Go / NodeJS	- NoSQL/SQL/RabbitMQ
- AWS	- Docker	- HTML/CSS
- Pulumi/Terraform	- Kafka	- CI/CD Workflows
- Design Patterns	- AWS Bedrock	- AI/ML Integration

EDUCATION

MS in Information Systems	Northeastern University	2024
BE in Information Science and Engineering	Visvesvaraya Technological University	2019

NOTABLE PROJECTS

- Built a multi-agent travel planning system** on AWS Bedrock with Supervisor, Collaborator, and Helper Agents using Claude and Nova models, deployed through API Gateway with real-time recommendation synthesis, **reducing travel planning time by 80%**.
- Engineered a concurrent text normalization system** with multi-threaded backend architecture leveraging AWS Bedrock LLM, implementing intelligent grammar correction with batch processing capabilities, **processing 10,000+ documents simultaneously with 95% accuracy**.